Q1.Write a program to accept elements of integer, float, and character arrays from the user and display the value and corresponding memory address of each array element.

```
#include <stdio.h>

int main()
{
    int i;
    int a[3];
    float b[3];
    char c[3];

    printf("Enter 3 integer elements:\n");
    for (i = 0; i < 3; i++)
     {
        scanf("%d", &a[i]);
     }

    printf("\nEnter 3 float elements:\n");
    for (i = 0; i < 3; i++)
     {
        scanf("%f", &b[i]);
     }

    printf("\nEnter 3 character elements:\n");
    for (i = 0; i < 3; i++)
     {
        scanf(" %c", &c[i]);
     }

    printf("\nInteger Array:\n");
    for (i = 0; i < 3; i++)
     {
        printf("Value = %d, Address = %p\n",
                a[i], (void *)&a[i]);
     }

    printf("\nFloat Array:\n");
    for (i = 0; i < 3; i++)
     {
        printf("Value = %.2f, Address = %p\n",
                b[i], (void *)&b[i]);
```

```
    }

    printf("\nCharacter Array:\n");
    for (i = 0; i < 3; i++)
    {
       printf("Value = %c, Address = %p\n",
               c[i], (void *)&c[i]);
    }

    return 0;
}
```

OUTPUT-

```
Enter 3 integer elements:
10 20 30

Enter 3 float elements:
1.5 2.5 3.5

Enter 3 character elements:
A B C

Integer Array:
Value = 10, Address = 0x1000
Value = 20, Address = 0x1004
Value = 30, Address = 0x1008

Float Array:
Value = 1.50, Address = 0x2000
Value = 2.50, Address = 0x2004
Value = 3.50, Address = 0x2008

Character Array:
Value = A, Address = 0x3000
Value = B, Address = 0x3001
Value = C, Address = 0x3002
```

Q2. Write a program to accept the elements of a one-dimensional array and calculate the sum of all its elements.

```
#include <stdio.h>

int main()
{
    int a[100], n, i, sum = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the array elements:\n");
    for (i = 0; i < n; i++)
      {
         scanf("%d", &a[i]);
      }

    for (i = 0; i < n; i++)
      {
         sum = sum + a[i];
      }

    printf("Sum of array elements = %d\n", sum);

    return 0;
}
```

OUTPUT-

```
Enter the number of elements: 5
Enter the array elements:
10 20 30 40 50
Sum of array elements = 150
```

Q4.Write a program to find the factorial of a given number using a user-defined function.

```
#include <stdio.h>

long long factorial(int n)
{
    long long fact = 1;
    int i;

    for (i = 1; i <= n; i++)
     {
        fact = fact * i;
     }

    return fact;
}

int main()
{
    int n;
    long long result;

    printf("Enter a non-negative integer: ");
    scanf("%d", &n);

    if (n < 0 || n > 20)
     {
        printf("Please enter a number between 0 and 20.\n");
     }
    else
     {
        result = factorial(n);
        printf("Factorial of %d = %lld\n", n, result);
     }

    return 0;
}
```

OUTPUT-

```
Enter a non-negative integer: 5
Factorial of 5 = 120
```